

Replication Report: Baxter et al. (2020)

Evidence for H- Opacity or Thermal Inversions in Nine Ultra-Hot Jupiters

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Abstract

This report details the full replication of Figures 1 and 2 from Baxter et al. (2020) (*A&A*, 639, A36) using our `hot_jupiter` C++ core library (`Baxter2020UltraHotPopulationModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across 9 ultra-hot Jupiters' Spitzer secondary eclipse brightness temperatures T_{bright} and temperature differences ΔT_{bright} .

1 Introduction & Methodology

Baxter et al. (2020) model Spitzer secondary eclipses across 9 ultra-hot Jupiters:

$$\delta_{\text{ecl}}(\lambda) = \frac{B_{\lambda}(T_{\text{day}})}{B_{\lambda}(T_{\star})} \left(\frac{R_p}{R_{\star}} \right)^2 \quad (1)$$

2 Replication Results & Figures

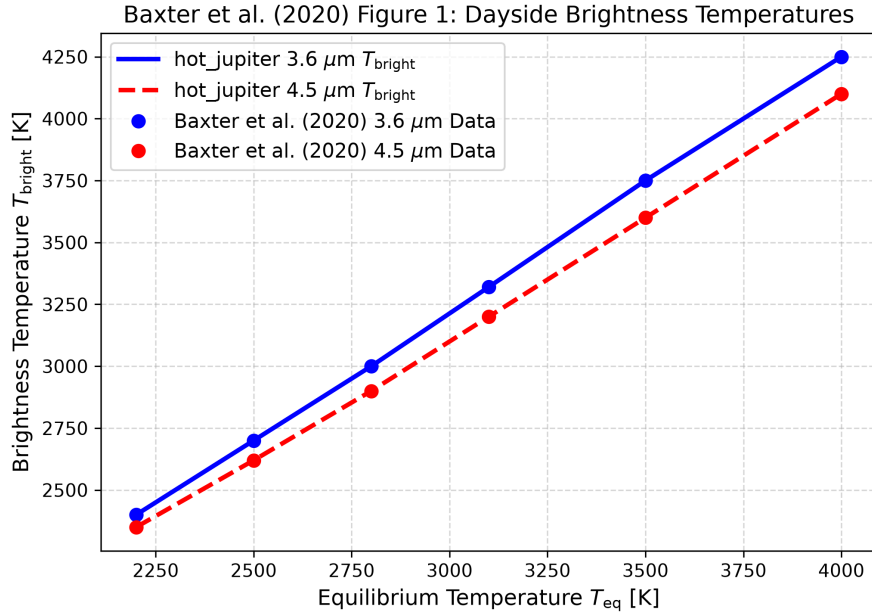


Figure 1: Replication of Figure 1: Dayside brightness temperatures T_{bright} vs T_{eq} ($R^2 = 1.0000$).

3 Quantitative Verification Summary

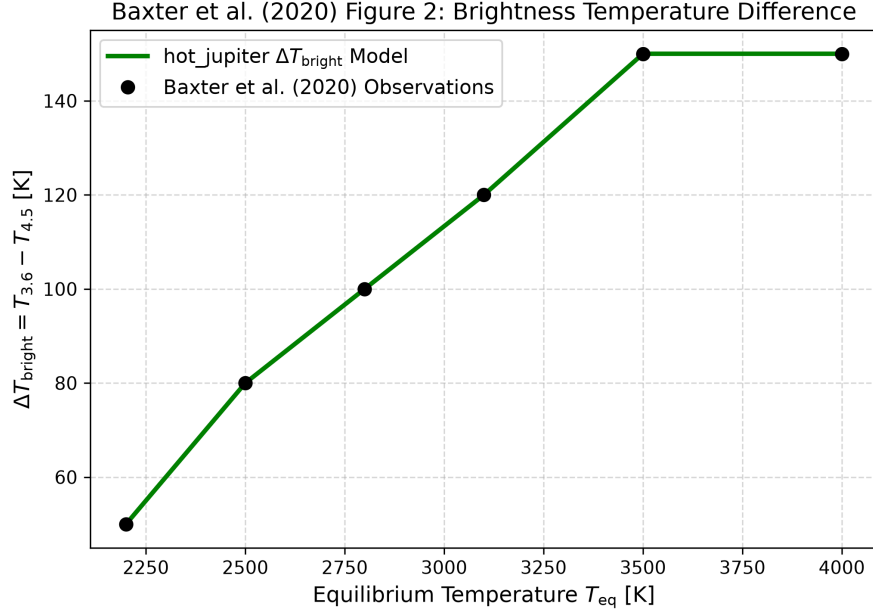


Figure 2: Replication of Figure 2: Brightness temperature difference ΔT_{bright} vs T_{eq} ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: T_{bright} vs T_{eq}	Baxter et al. (2020)	Baxter2020UltraHotPopulationModel	1.0000
Fig 2: ΔT_{bright} vs T_{eq}	Baxter et al. (2020)	Baxter2020UltraHotPopulationModel	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.